FACILITATOR

PRINT & PREP CHECKLIST

Print this pack once, then duplicate the pages marked **per team**. Everything is black-and-white friendly. Total prep time: about 25 minutes for a class of 24 (6 teams of 4).

PAGE	ITEM	COPIES	PREP
2	Incident Briefing	1 per team	Hand out at start
3	Team Roster + Role Cards	1 per team	Cut the 5 role cards apart
4	Station A - Parts Requisition Form	1 per team	Envelope A
5-6	Station A - Component Cards (10)	1 set per team	Cut apart, shuffle, Envelope A
7	Station B - Memory Dump	1 per team	Envelope B
8	Station B - Binary + ASCII Reference	1 per team	Envelope B
9	Station C - Network Map + Host Table	1 per team	Envelope C
10	Station C - Subnet + Ports Reference	1 per team	Envelope C
11	Station D - Ransom Note	1 per team	Envelope D
12-13	Station D - Cipher Wheel discs	1 per team	Cut out, join with a brad
14	Recovery Console (final worksheet)	1 per team	Give at the start
15	Hint Cards (9)	1 set per team	Cut apart, facilitator keeps
16	Bonus - Phishing Triage	fast finishers	Hold in reserve
17	Exit Ticket	1 per student	Debrief
18	Completion Certificate	1 per student	Optional

ALSO GATHER (optional):

4 x four-digit combination locks, 1 lockbox or toolbox, 5 manila envelopes per team, brads/paper fasteners (1 per team), scissors, a visible timer, and a small prize for the box. No locks? Seal each envelope and write the code on the outside — teams may only open an envelope once they can tell you the code. Or use the Google Form version in the Facilitator Guide.

BRIEFING

INCIDENT TICKET #4829

PRIORITY 1 — SEV-1 — ACTIVE INCIDENT

07:42 — Gulf Coast Data Services calls the help desk. Every workstation in their office is showing the same message: a red screen, a countdown, and a ransom demand signed **BLACKOUT**. The primary server is locked. Payroll runs today.

07:51 — Their only IT tech quit two weeks ago. He left no passwords, no diagrams, and no notes — just four sealed envelopes in a desk drawer marked **A**, **B**, **C**, and **D**, and a locked recovery box.

07:58 — That is where you come in. You are the incident response team. You have **45 minutes** before BLACKOUT wipes the drives.

YOUR MISSION

Each envelope contains a puzzle left behind by the old tech. Each puzzle produces a **four-digit station code**. Log all four codes on your **Recovery Console** worksheet, then follow the recovery protocol printed on it to assemble the **MASTER CODE** that opens the recovery box.

RULES OF ENGAGEMENT

1. Work the envelopes in any order. Split up — you do not all have to look at the same page.
2. Everything you need is in the envelopes. No phones, no searching. The reference cards are your documentation.
3. Nothing counts until the Documentation Lead writes it on the Recovery Console.
4. You may call for a **HINT** three times. Raise a hand and say which station.
5. Do not force, tear, or dismantle anything. Every lock opens with information, not muscle.

Read the ransom note last if you want to sleep tonight. — Help Desk Supervisor

TEAM SETUP

ROSTER & ROLE CARDS

Write your team name and assign every role before you open a single envelope. If your team has four people, the Timekeeper also takes Comms. If you have three, the Incident Commander doubles up.

TEAM NAME		START TIME		STOP TIME	

- CUT THE FIVE ROLE CARDS APART -

INCIDENT COMMANDER	Keeps the team moving. Decides what the team attacks next and who works on what. The only person who may call for a hint. Does not have to be the person who knows the most.	NAME: _____
DOCUMENTATION LEAD	Owns the Recovery Console. Writes down every code the moment it is confirmed. If it is not written down, it did not happen.	NAME: _____
ANALYST	Leads the math: binary conversion, IP addressing, and the cipher wheel. Shows the work so a teammate can check it.	NAME: _____
HARDWARE SPECIALIST	Leads component identification. Keeps the cards laid out face up and organized so nothing gets lost under a folder.	NAME: _____
TIMEKEEPER / COMMS	Watches the clock and calls out the 30, 15, and 5 minute marks. Speaks to the facilitator on behalf of the team.	NAME: _____

Every role solves puzzles. The titles describe who is responsible, not who is allowed to think.

ENVELOPE A

PARTS REQUISITION FORM

The old tech left a purchase requisition, but he never wrote down the part names — only what each part **does**. Ten component cards are in this envelope. Match each requisition line to the correct card, then write that card's **ID digit** in the box.

LINE	FUNCTION REQUESTED	COMPONENT NAME	ID
R1	Executes program instructions. The system's decision-maker.		
R2	Volatile working space. Everything stored here is lost the instant power is removed.		
R3	Non-volatile storage. Holds the operating system while the machine is powered off.		
R4	Converts AC from the wall outlet into the DC voltage rails the system needs.		

STATION A CODE — read the four ID digits top to bottom, R1 to R4.

CODE A = | | | | |

Six of the ten cards are decoys. That is on purpose — real requisitions arrive with parts you did not order.

ENVELOPE A

COMPONENT CARDS 1 of 2

- CUT APART, THEN SHUFFLE -

MOTHERBOARD

Large printed circuit board. ATX form factor, 30.5 x 24.4 cm. Chipset, DIMM slots, PCIe slots, 24-pin header, SATA headers, rear I/O shield.

ID 1

NETWORK INTERFACE CARD

PCIe x1 expansion card. Single RJ-45 port, 10/100/1000 Mbps, two link LEDs, burned-in 48-bit hardware address.

ID 2

MEMORY MODULE

DDR4 DIMM. 8 GB, 3200 MT/s, 288 pins, off-center notch, low-profile aluminum heat spreader. Sold in matched pairs.

ID 3

SOLID-STATE DRIVE

2.5 inch, SATA III, 500 GB. NAND flash, no moving parts, 7 mm z-height, data + power connectors on one edge.

ID 4

OPTICAL DRIVE

5.25 inch bay device. Motorized tray, 650 nm laser assembly, spindle motor, eject button and pinhole. Increasingly rare in new builds.

ID 5

GRAPHICS CARD

PCIe x16, dual-slot. Onboard GDDR6, dedicated cooler, 8-pin auxiliary power connector, HDMI and DisplayPort outputs.

ID 6

ENVELOPE A

COMPONENT CARDS 2 of 2

- CUT APART, THEN SHUFFLE -

PROCESSOR

Square package roughly 37 x 37 mm. LGA1700, 1700 contact pads, integrated heat spreader, gold triangle in one corner for orientation.

ID 7

CPU COOLER

Finned aluminum tower with four copper heat pipes and a 92 mm PWM fan. Mounts to the retention bracket. Requires thermal compound.

ID 8

POWER SUPPLY UNIT

Steel enclosure, 150 x 140 x 86 mm. 650 W, 80 PLUS Gold, 120 mm fan, IEC C14 inlet, modular cable bundle: 24-pin, EPS, PCIe, SATA.

ID 9

MONITOR

27 inch IPS panel, 1920 x 1080, 75 Hz. External peripheral. HDMI in, DisplayPort in, VESA 100 mm mount, tilt stand.

ID 0

SPARE / BLANK CARDS — write your own components here if you want to extend the puzzle, or leave blank as extra decoys.

ENVELOPE B

MEMORY DUMP 0x00 — 0x05

Before BLACKOUT locked the console, the server flushed six bytes of memory to the crash log. The old tech believed these six bytes spell the **recovery command**.

OFFSET	BINARY (8 BITS)	DECIMAL	CHARACTER
0x00	0 1 0 1 0 0 1 0		
0x01	0 1 0 0 0 1 0 1		
0x02	0 1 0 0 0 0 1 0		
0x03	0 1 0 0 1 1 1 1		
0x04	0 1 0 0 1 1 1 1		
0x05	0 1 0 1 0 1 0 0		

THE RECOVERY COMMAND

_____ _____ _____ _____ _____ _____

STATION B CODE — take the **decimal value** of the **first** character, then the **decimal value** of the **last** character, and write them side by side.

CODE B = | | | | |

REFERENCE B

BINARY & ASCII DECODE CARD

PLACE VALUES — write these above any byte, then add the columns that hold a 1.

128	64	32	16	8	4	2	1

WORKED EXAMPLE

`0 1 0 0 1 0 0 0` → `64 + 8 = 72` → `the letter H`

ASCII — UPPERCASE LETTERS

A	B	C	D	E	F	G	H	I	J	K	L	M
65	66	67	68	69	70	71	72	73	74	75	76	77
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
78	79	80	81	82	83	84	85	86	87	88	89	90

WHY IT MATTERS

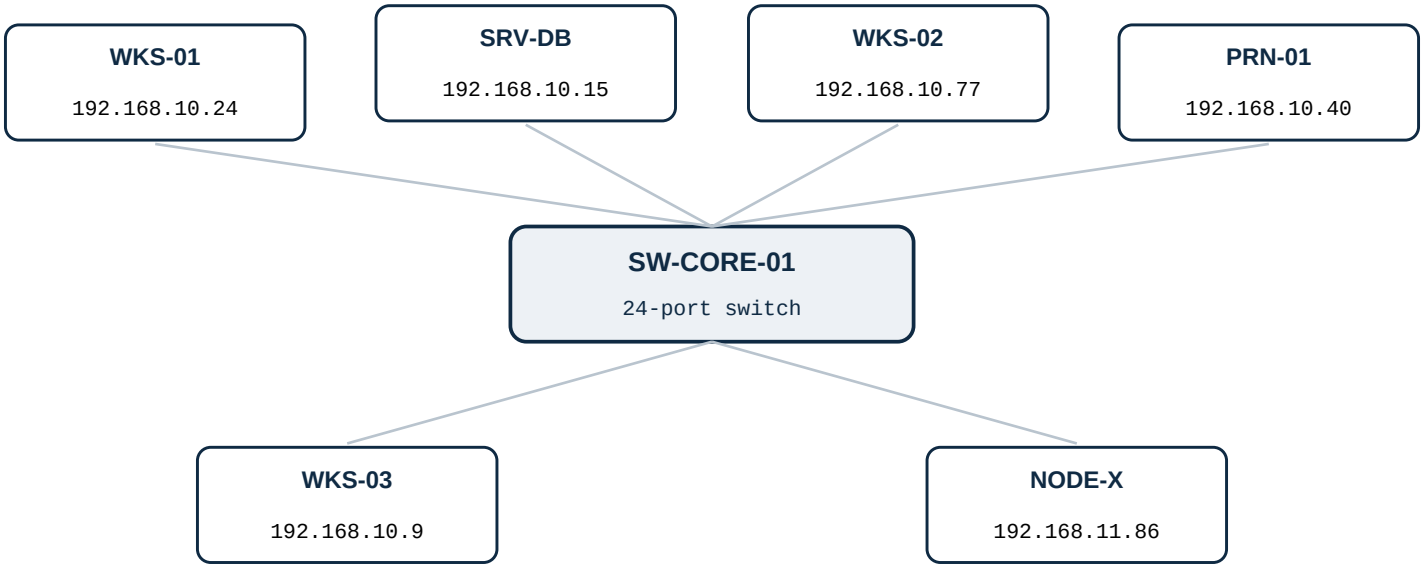
A computer stores no letters — only voltages, read as 1s and 0s. ASCII is the agreement that the number 82 means the letter R. Change the agreement and the same bits become something else entirely. This is the whole idea behind character encoding, and it is why a file opened with the wrong encoding turns into garbage.

SCRATCH SPACE

ENVELOPE C

NETWORK MAP & HOST TABLE

This is the client network as the old tech last documented it. Every legitimate device sits on **192.168.10.0 /24**. One device on this map does not belong. Find it.



HOSTNAME	IP ADDRESS	SUBNET MASK	ROLE	ON THE /24?
WKS-01	192.168.10.24	255.255.255.0	Front desk workstation	
SRV-DB	192.168.10.15	255.255.255.0	Database server	
WKS-02	192.168.10.77	255.255.255.0	Accounting workstation	
PRN-01	192.168.10.40	255.255.255.0	Network printer	
WKS-03	192.168.10.9	255.255.255.0	Shop floor workstation	
NODE-X	192.168.11.86	255.255.255.0	UNDOCUMENTED	

TRAFFIC LOG EXCERPT

```
08:03:11 ALLOW <rogue host> -> 203.0.113.44 dst-port ???? 1.4 GB out
08:03:11 NOTE session is plaintext; credentials visible in capture
08:03:12 NOTE legacy file-transfer protocol, separate control and data channels
```

STATION C CODE

Part 1 — the **host octet** (last number) of the device NOT on the /24 network. **Part 2** — the **control port** of the protocol in the traffic log. Part 1 first, then Part 2.

CODE C = | | | | |

REFERENCE C

SUBNET & PORT CARD

READING AN IPv4 ADDRESS

```
192 . 168 . 10 . 24
 |       |      |     |
NETWORK PORTION (/24)   HOST
```

A **/24** (mask 255.255.255.0) means the **first three octets identify the network** and only the fourth octet identifies the individual host. Two devices can talk directly to each other only if all three network octets match exactly. If they do not match, traffic must be routed — and an undocumented device on a different network sitting on your switch is a red flag, not a typo.

QUICK CHECK — compare one column at a time:

```
192.168.10.24  and  192.168.10.77  →  same network
192.168.10.24  and  192.168.11.86  →  DIFFERENT network
```

COMMON PORTS AND PROTOCOLS

PORT	PROTOCOL	WHAT IT DOES	ENCRYPTED?
20	FTP (data)	File transfer - data channel	No
21	FTP (control)	File transfer - commands and login	No
22	SSH / SFTP	Secure remote shell and file transfer	Yes
23	Telnet	Legacy remote shell	No
25	SMTP	Sending mail between servers	No
53	DNS	Resolves names to IP addresses	No
67 / 68	DHCP	Hands out IP addresses automatically	No
80	HTTP	Web traffic	No
110	POP3	Retrieving mail	No
143	IMAP	Retrieving mail, server-side	No
443	HTTPS	Web traffic over TLS	Yes
3389	RDP	Windows remote desktop	Yes

"Legacy" and "unencrypted" are the two words that should make an analyst sit up. Anything in the No column can be read by anyone sitting between the two endpoints.

ENVELOPE D

THE RANSOM NOTE

This text was printed on every screen in the building. It is enciphered. BLACKOUT is arrogant enough to have left the shift value in the signature.

DTZW XDXYJRX FWJ RNSJ.

YT WJXYTWJ, FUUQD YMJ UFYHM.

TAJWWNIJ HTIJ:

YMWJJ TSJ SNSJ YBT

— BLACKOUT

shift = the number of letters in the word CYBER

DECODE IT HERE

_____

_____

_____

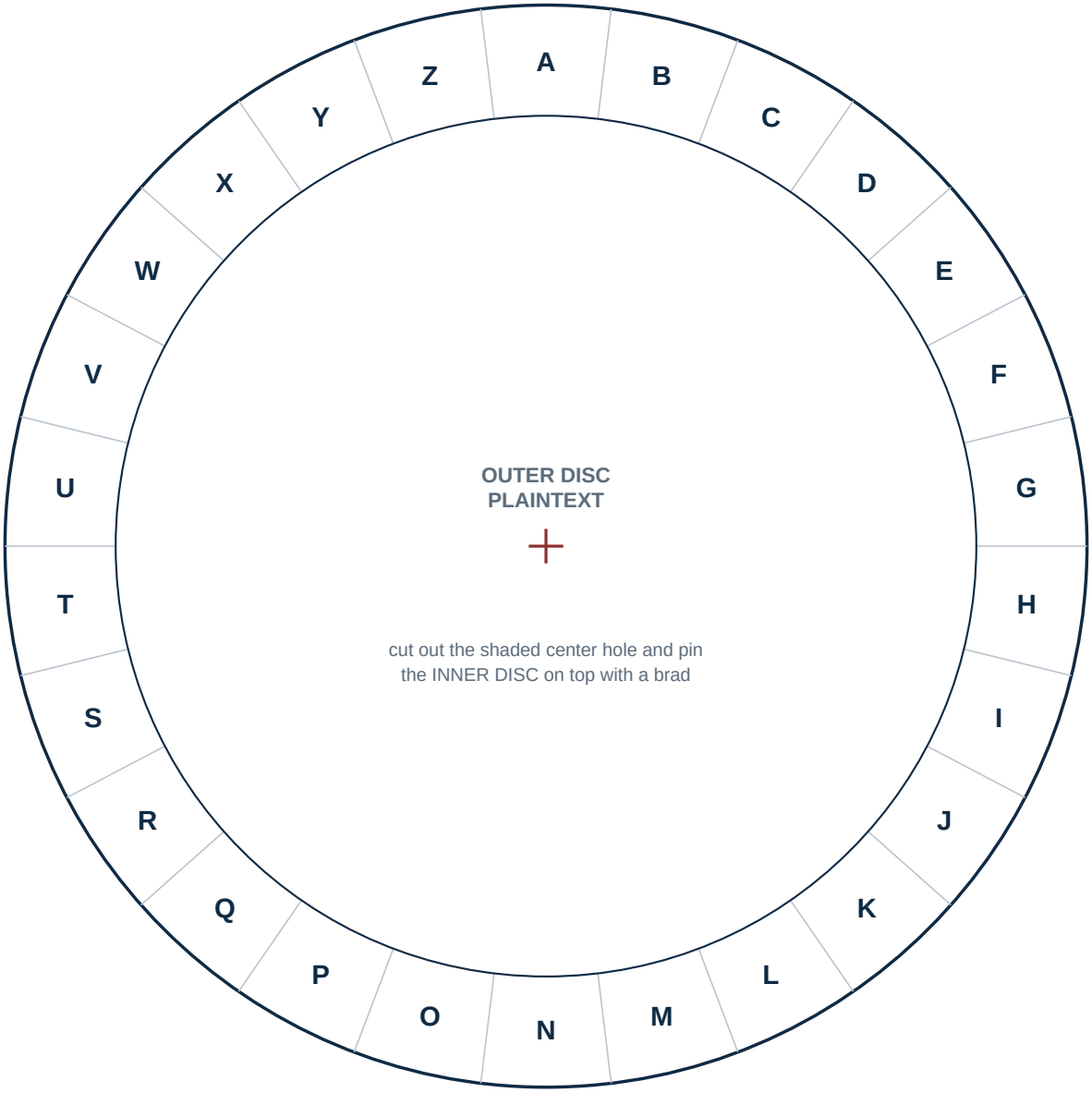
STATION D CODE — the note spells the override code out in words. Write it as four digits. (If your station uses a five-letter word lock, the note also names the action you must take — that word is your lock code.)

CODE D = | | | | |

TOOL D

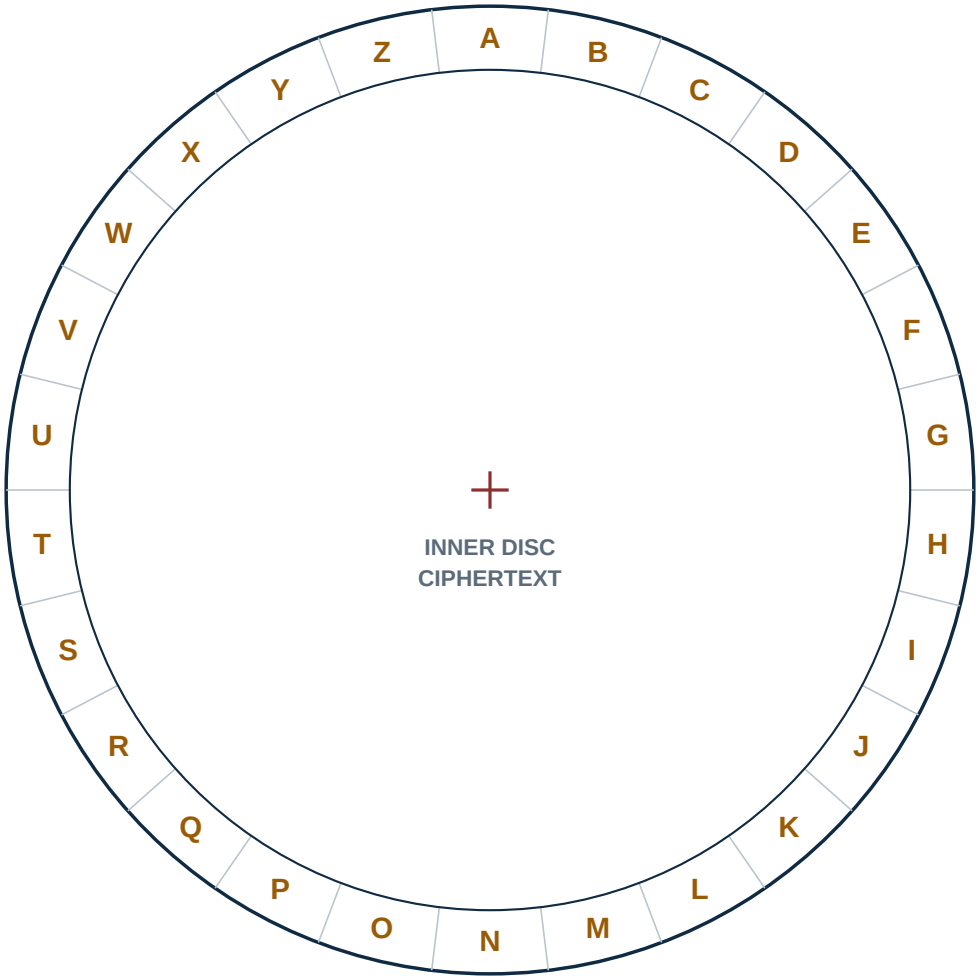
CIPHER WHEEL — OUTER DISC

Cut out both discs (this page and the next). Push a brad through the red cross at the center of each, inner disc on top. Now you can rotate one alphabet against the other.



TOOL D

CIPHER WHEEL — INNER DISC



HOW TO USE IT

1. Work out the shift value from the signature on the ransom note.
2. A Caesar cipher shifts *forward* to encrypt. With a shift of 3, plaintext A becomes ciphertext D. So line up **OUTER A** with the ciphertext letter that is that many steps ahead on the **INNER** ring.
3. To decode: find each ciphertext letter on the **INNER** ring and read the letter directly outside it on the **OUTER** ring.
4. Once the wheel is aligned, it stays aligned for the whole message. Do not spin it again.

WHY THIS MATTERS: the Caesar cipher is roughly 2,000 years old and a modern laptop breaks it in under a millisecond — there are only 25 possible keys. It survives today only as a teaching tool and as ROT13 on internet forums. Every real cipher you will meet later solves the same problem this one does; it just uses a keyspace too large to walk through by hand.

FINAL

RECOVERY CONSOLE

Log every station code the moment you confirm it. When all four are logged, run the recovery protocol at the bottom to assemble the MASTER CODE.

STATION	SOURCE	CODE (4 DIGITS)	THIRD DIGIT	CONFIRMED BY
A	Parts Requisition			
B	Memory Dump			
C	Network Map			
D	Ransom Note			

RECOVERY PROTOCOL

The recovery box does not accept any single station code. Take the **THIRD DIGIT FROM THE LEFT** of each station code and write them in order A, B, C, D.

`EXAMPLE:  1 2 3 4  →  third digit is 3`

MASTER CODE

Bring this sheet to the facilitator with all four station codes filled in. An incident is not resolved until it is documented — in this room and in every job you will ever hold.

FACILITATOR

HINT CARDS — CUT APART

Hand these out one at a time when a team calls a hint. Give the level-1 card first; only give level 2 if they are still stuck after two or three minutes.

A - LEVEL 1

Ignore the specs for a second. Read only the requisition line and ask: what job does this part do? Then find the card that could do that job.

A - LEVEL 2

R1 is a square chip about the size of a postage stamp. R4 is the only card with a wattage rating.

A - LEVEL 3

Order matters. Read the ID digits R1, R2, R3, R4 - top to bottom, not sorted.

B - LEVEL 1

Write 128 64 32 16 8 4 2 1 above the eight bits. Add up only the columns that have a 1 under them.

B - LEVEL 2

Those decimal numbers are ASCII codes. Use the letter table to turn each one into a character first.

B - LEVEL 3

The word is REBOOT. Now go back to your decimal column: what number is under R, and what is under T?

C - LEVEL 1

A /24 means the first THREE octets must match. Compare the addresses one column at a time.

C - LEVEL 2

The rogue device is NODE-X. Now: legacy, unencrypted, separate control and data channels. It is not SSH.

D - LEVEL 1

Count the letters in the word CYBER. That number is your shift. Set the wheel and leave it alone.

D - LEVEL 2

Ciphertext letters live on the INNER ring. Find each one there and read outward to the OUTER ring.

D - LEVEL 3

The last line spells four digits out in English words. Write the digits, not the words.

FINAL

The console wants the third digit from the LEFT of each code. In 7 3 4 9 that would be 4.

BONUS

PHISHING TRIAGE — FAST FINISHERS

You unlocked the box early. Before the client goes back online, triage the five messages sitting in the help desk queue. Mark each **PHISH** or **SAFE**, and name the single strongest signal that decided it.

#	MESSAGE	PHISH / SAFE	STRONGEST SIGNAL
1	FROM: it-support@gulfcoastdata.co | SUBJ: Mandatory password reset "Your account expires in 2 hours. Click here to keep access."		
2	FROM: no-reply@ups.com | SUBJ: Delivery scheduled for Tuesday "Tracking 1Z8842. No action needed." (You ordered a switch last week.)		
3	FROM: ceo.gulfcoast@gmail.com | SUBJ: quick favor, in a meeting "Need you to buy 4 gift cards for a client. Send me the codes. Keep this between us."		
4	FROM: billing@microsoft-secure-login.net | SUBJ: Invoice 88213 overdue "Open the attached .zip to view your invoice and avoid suspension."		
5	FROM: jsmith@gulfcoastdata.com | SUBJ: Re: printer on the shop floor "Still jamming. Can someone come look at it today?" (Reply in an existing thread you started.)		

THEN ANSWER: BLACKOUT got in somehow. Based on these five messages, write one sentence naming the most likely entry point, and one sentence proposing a control that would have stopped it.

DEBRIEF

EXIT TICKET

NAME _____ TEAM _____

1. Which station did your team solve fastest, and what made it click?

2. Which station stalled you the longest? What finally moved it?

3. Name one thing you know now that you did not know at 8 a.m. today.

4. Your team had five roles. Which one did you actually end up doing most, and was it the one on your card?

5. Real incident response looks a lot like this: incomplete documentation, time pressure, and a team you just met. What is one habit you want to bring into this program because of that?

CERTIFICATE OF COMPLETION

SYSTEM LOCKDOWN

INCIDENT #4829 — RESOLVED

This certifies that

_____

served as a member of the incident response team, recovered a compromised client network under time pressure, and returned Gulf Coast Data Services to operation.

DEMONSTRATED: hardware component identification • binary and ASCII conversion • IPv4 addressing and subnet logic • port and protocol recognition • basic cryptanalysis • incident documentation • collaborative problem solving under pressure

_____

TIME TO ESCAPE **INSTRUCTOR**